Plasma Stability

Plasma Dynamics:

To analyze a non-equilibrium problem, we have to retain the momentum terms.

An ideal MHD is one with no resistivity, This is a good approximation for high frequencies.

An ideal MHD gives us: $\vec{E} + \vec{v} \times \vec{B} = 0$; $\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$

This gives us: $\nabla \times (\vec{v} \times \vec{B}) = +\frac{\partial \vec{B}}{\partial t}$ → changes in $\vec{B}$ are completely determined by plasma flow.

Flux Conservation

Consider a closed contour C with a spawning surface S.

Flux linked by C is: $\phi = \int_S \vec{B} \cdot d\vec{s}$

If C and S move with the fluid, rate of change of ϕ is given by two terms:

$$\frac{D\phi}{Dt} = \frac{\partial}{\partial t} \int_S \vec{B} \cdot d\vec{s} = \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s} + \int_S \vec{B} \cdot \frac{\partial s}{\partial t} = \int_S \frac{\partial \vec{B}}{\partial t} \cdot d\vec{s} + \oint_C \vec{B} \cdot (\vec{v} \times d\vec{l}) = -\oint_C (\vec{E} + \vec{v} \times \vec{B}) \cdot dl = 0$$

This implies that flux conservation is only possible when we have ideal Ohm's law.

Field-Line Motion

We can think of a field line as an intersection of two flux surfaces. These surfaces are tangential to the field everywhere.

We can say that: no. of field lines through any surface is constant and that a field line that starts as a field line, remains one.

MHD Stability

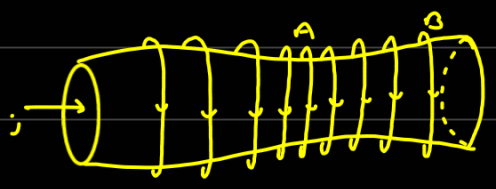
We have seen how we can find MHD equilibria, however one problem with this is that the equilibria may be unstable.

An unstable equilibrium is characterized by: $\frac{d^2\{\omega_{pot}\}}{dx^2} < 0$ → Notice the similarities to the maxima condition.

MHD has the potential energy: Magnetic + Kinetic Pressure.

An unstable equilibrium will remain only for a short time. If we can find any perturbation that lowers P.E, we have an unstable eq^m.

Let's look at an example perturbation: The sausage instability.



The question we are asking is: Is this eq^m stable?

We simplify this picture by taking the current to all flow in a skin.

We have pressure and tension forces. Now, these forces are $\propto B$ and the tension force $\propto \frac{1}{r}$

Let's have a look, At A: $B_\theta \uparrow$, $\frac{1}{r} \uparrow$, Thus pressure and tension forces increase which cannot be balanced by kinetic pressure.

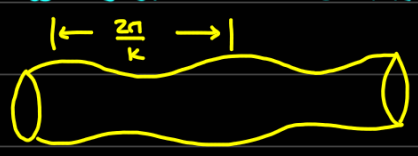
At B, its the opposite.

Its very simple now to see that this equilibrium will dissipate. The outward part will grow while the inward part will shrink and this will continue.

Thus, the sausage instability is unstable.

General Perturbations

Look at waves of the mathematical form: $\exp[i(kz+m\theta)]$



Here, the kz contributes to a wave-like behaviour in z while the $m\theta$ contributes to the wavelike behaviour in θ, making it snake-like.

If $m=0$ we have a sausage instability. Similarly, $m=1 \rightarrow$ kink, $m=2 \rightarrow$ higher order kink.

General Principles Governing Instabilities

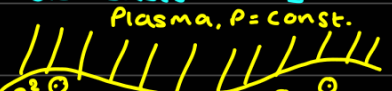
1.) Instabilities try not to bend field lines. Bending these will require energy.

This implies that the perturbation lies along the magnetic field (i.e constant surfaces lie along $\vec{B}$)

2.) Instabilities occur when a "heavier" fluid is supported by a "lighter" fluid. This is the case with supporting a plasma using a mag. field. This is similar to how an inverted glass of water falls out of a glass as its no longer supported by air pressure.

3.) Generally occur when $|B|$ decreases away from the plasma region:

$B_A^2 < B_B^2$



You can see how the force closer to the

$2\mu_0 \quad 2\mu_0$

plasma is more than the force away, thus increasing the instability.

4.) Take place when field line curvature is towards the plasma. This is the same as (3) as $\nabla \times \vec{B} = 0$ in a vacuum.

An Analysis of Pinches

Consider this: θ-pinch, $|B| =$ const. ⟶ No field line curvature. Neutral stability.

z-pinch, $\nabla|B|$ away from plasma outside ⟶ Bad curvature (Towards plasma). Instability.

We can think of good curvature as lesser and usually, closer to eq^n. It is quite difficult to get a curvature to be good everywhere.

We can make do with a curvature that is good on average. This works because then the plasma will have to bend the field line a fair bit which is not preferred.

Another stabilizing method is to introduce a sheared field. This would make field line bending inevitable, increasing stability.